

16-bit unsigned sequential multiplier

تنظيم وبناء حاسب COE321/ 2

شعبة: M1

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Objective:

The objective of this project is to design and simulate a **16-bit unsigned sequential multiplier** using Logisim Classic. The goal is to understand binary multiplication using sequential addition and shifting for larger word sizes.

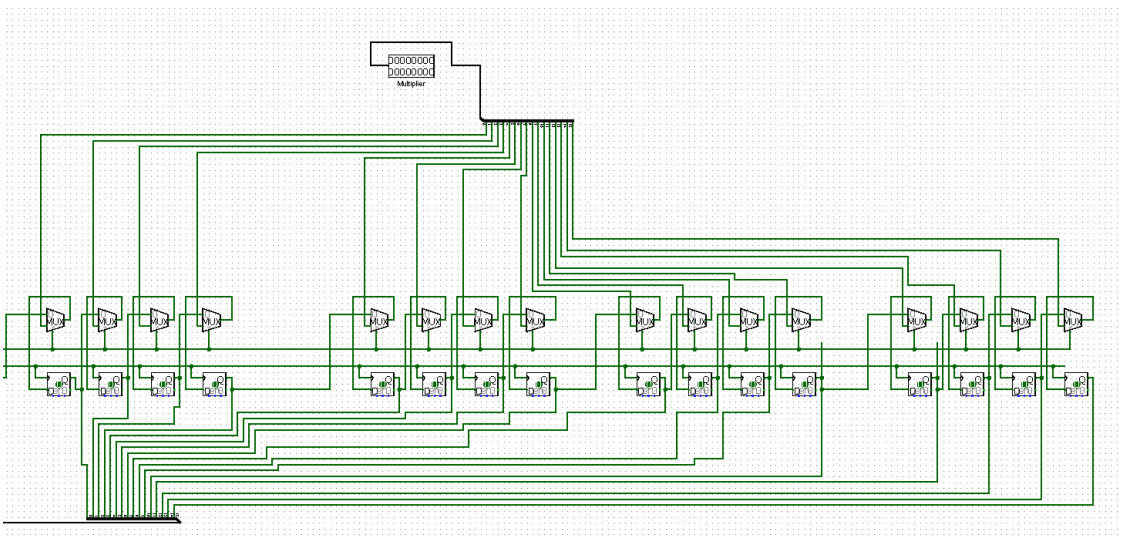
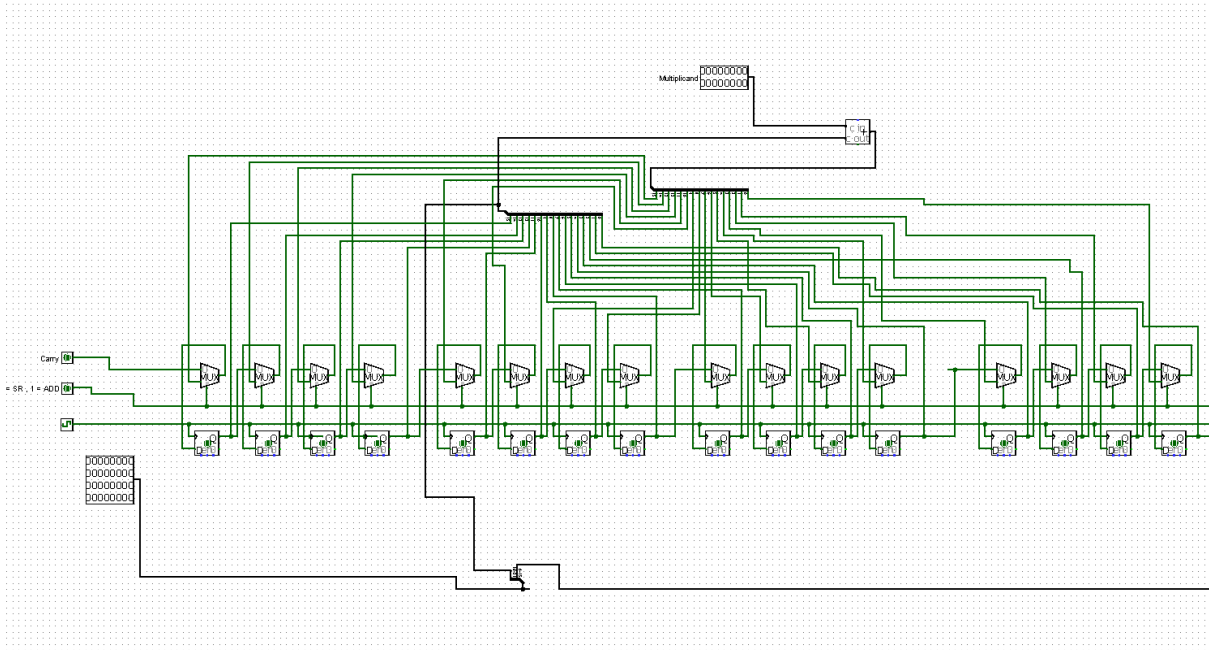
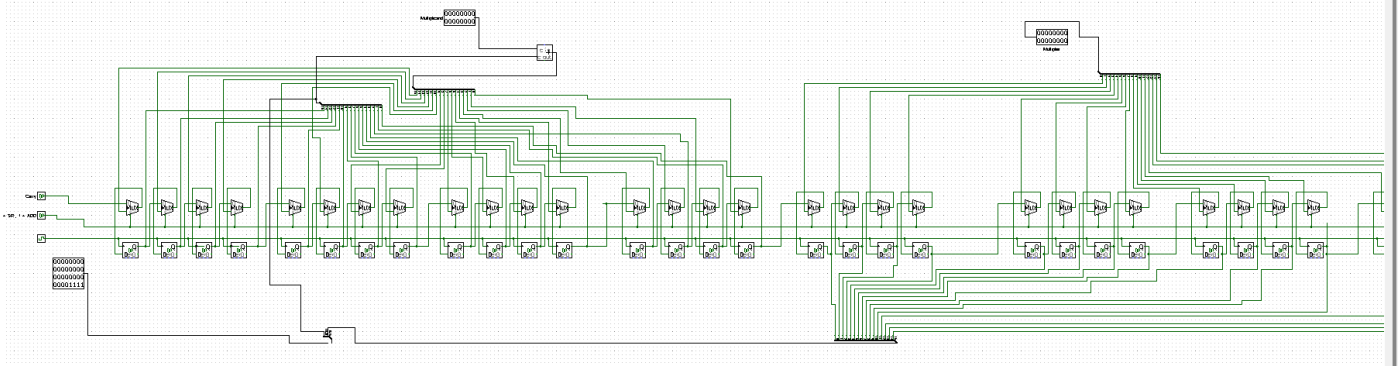
Design Description:

The circuit implements a **sequential multiplication algorithm**.

It multiplies two 16-bit unsigned binary numbers through a sequence of additions and right shifts.

The main components used in the design are:

- **Pins:** (multiplicand, multiplier, control signals, and product outputs)
- **Multiplexers:** used to select between shift input and add/load input for each bit in A and Q.
- **D Flip-Flops:** used as registers for A, Q, and Carry to store intermediate results.
- **Adder (16-bit):** performs $A + \text{multiplicand}$ during the ADD phase.
- **Splitters:** used to group and separate multi-bit buses for A, Q, the adder, and the 32-bit product.

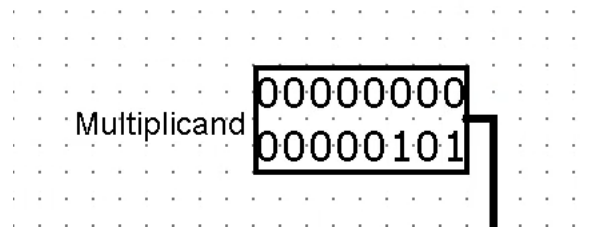


Simulation Results:

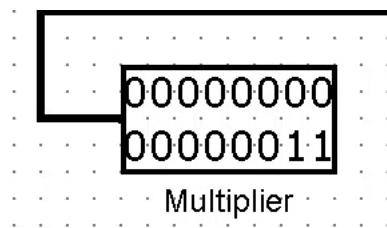
The circuit was tested with multiple input combinations to verify correct operation.

Example test case:

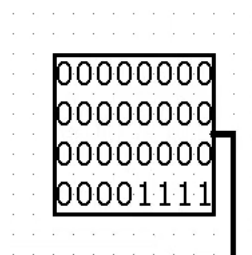
- **Input A (multiplicand)** = 0000 0000 0000 0101 (5)



- **Input B (multiplier)** = 0000 0000 0000 0011 (3)



- **Output (product)** = 0000 0000 0000 0000 0000 0000 0000 1111 (15)



The simulation results were correct, confirming that the design performs accurate 16-bit unsigned binary multiplication.

Conclusion:

The **16-bit unsigned sequential multiplier** was successfully designed and simulated using Logisim Classic.

The project demonstrates the process of binary multiplication using sequential addition, D flip-flops, multiplexers, and simple control logic.